



Distributed Image Reconstruction with Safe Multi-Agent Coordination

High-Density Swarm Control using
HOCBF-CLF and Auction Algorithms

Pranav Jain | University of Michigan



Introduction & Motivation



Goal: Physicalizing digital images using autonomous agents.



Challenge: Scaling decentralized decision-making for high-density swarms.



Innovation: Bridging digital representation with safety-critical coordination.

The Three-Stage Pipeline

1. Spatial Extraction: Farthest Point Sampling (FPS) of source images.

A horizontal orange rounded rectangle with a light pink arrow pointing downwards from its right side.

2. Task Allocation: Distributed Auction mechanism with color constraints.

A horizontal green rounded rectangle with a light grey arrow pointing downwards from its right side.

3. Safety Control: Decentralized HOCBF-CLF Trajectory Generation.

A horizontal blue rounded rectangle.

Image-to-Point Mapping

FPS Sampling: Ensures uniform point distribution.

Color Quantization: 3-bit binary mapping (7 active color classes).

Efficiency: Background filtering (White pixels > 0.95) reduces overhead.

BINARY COLOR MAPPING AND ROBOT INITIALIZATION CLASSES

Color ID	Red	Green	Blue	Resulting Color
1	0	0	0	Black
2	0	0	1	Blue
3	0	1	0	Green
4	0	1	1	Cyan
5	1	0	0	Red
6	1	0	1	Magenta
7	1	1	0	Yellow
8 (<i>Ignored</i>)	1	1	1	White (<i>Background</i>)

Distributed Task Allocation

Mechanism: Distributed Auction Algorithm.

Constraints: Robots only bid on targets matching their specific color ID.

Cost Function: Euclidean distance minimization ($O(N^2)$ complexity).

Control
Framework:
Unicycle
Dynamics & CLF

Model: Non-holonomic unicycle dynamics.

Convergence: Control Lyapunov Function (CLF) for goal attraction.

Nominal Control: Desired velocity proportional to distance error.

Safety: High-Order Control Barrier Functions

Barrier: Maintains minimum separation distance D_{safe} .

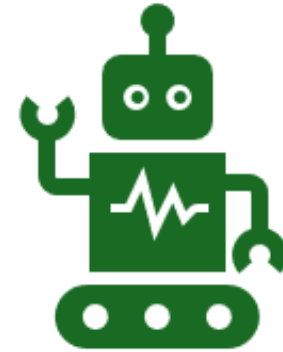
Adaptive Tuning: Local gamma relaxation allows for high-density packing at goal.

QP Solver: Ensures safety is prioritized over the nominal goal.

Deadlock Mitigation & Yielding

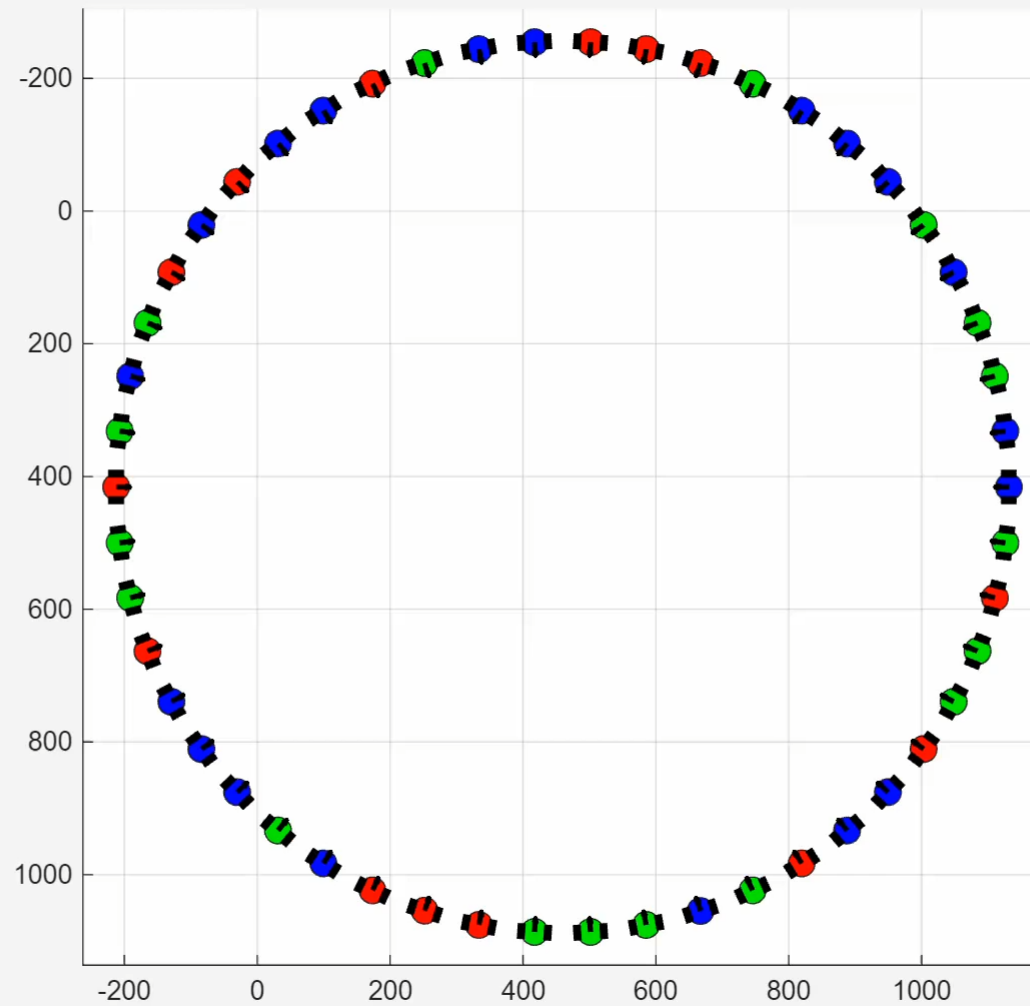


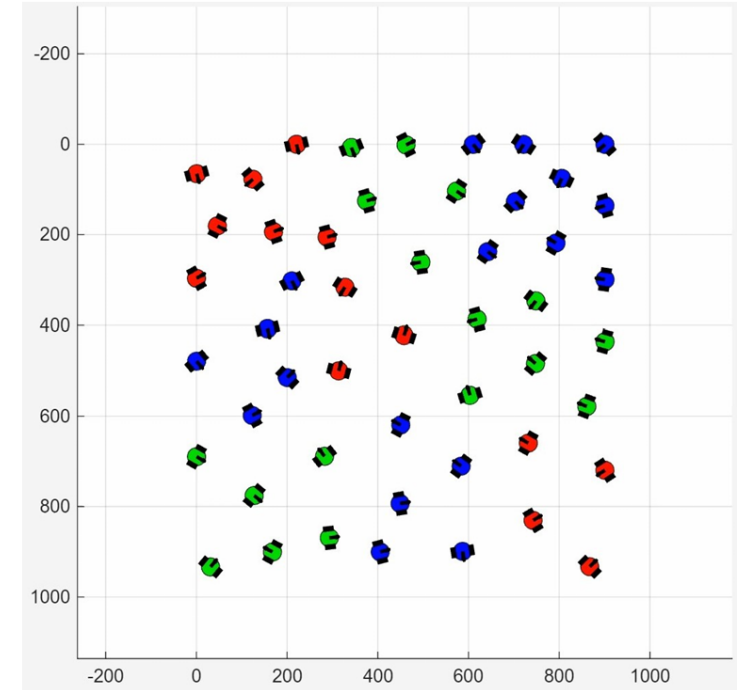
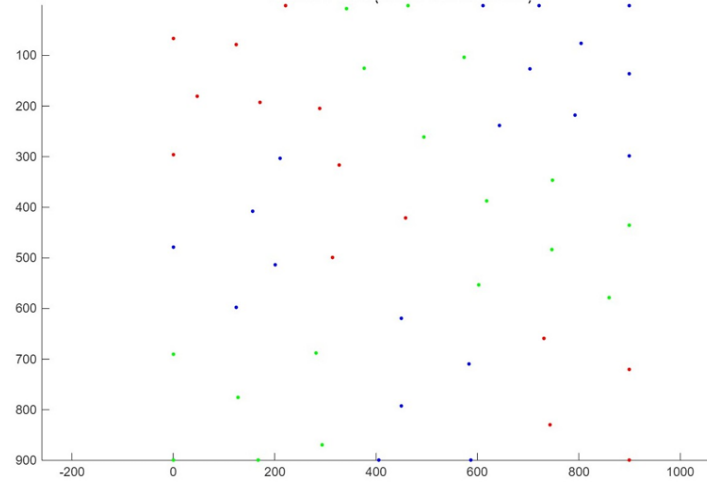
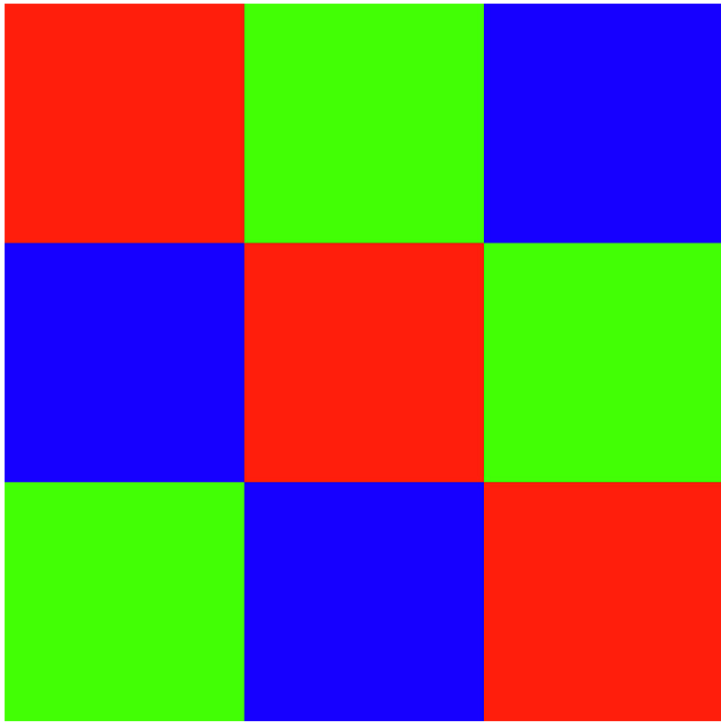
Sequential Activation: Populating the image center first to prevent blockages.



Directional Yielding: Active robots move orthogonally to avoid incoming neighbors.

Video





Simulation Results

- ✓ **Success:** Zero collisions across formations.
- ✗ **Trade-off 1:** Sequential release ensures safety but increases time.
- ✗ **Trade-off 2:** Oscillations due to competing barrier constraints
- ✗ **Trade-off 3:** Binarization simplifies allocation but loses color variations (e.g. orange to red).

Conclusion & Future Work



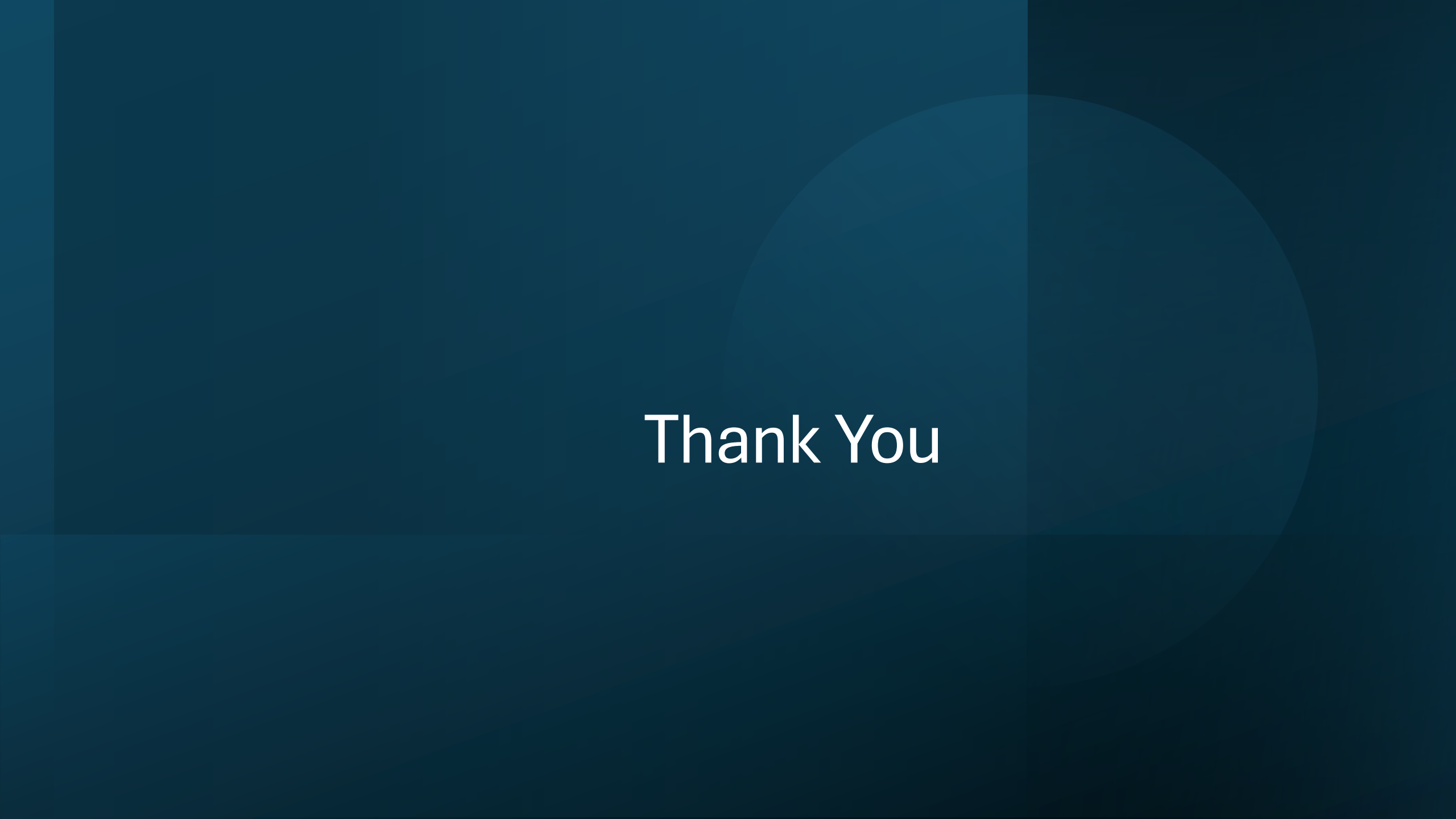
SUMMARY: INTEGRATED AUCTION-BASED TASKING WITH HOCBF-CLF.



FUTURE: DENSITY-AWARE RELEASE POLICIES AND SMOOTHING OF OSCILLATORY INPUTS.



EXPANSION: MOVING FROM 2D PLANES TO 3D UAV COORDINATION.



Thank You